



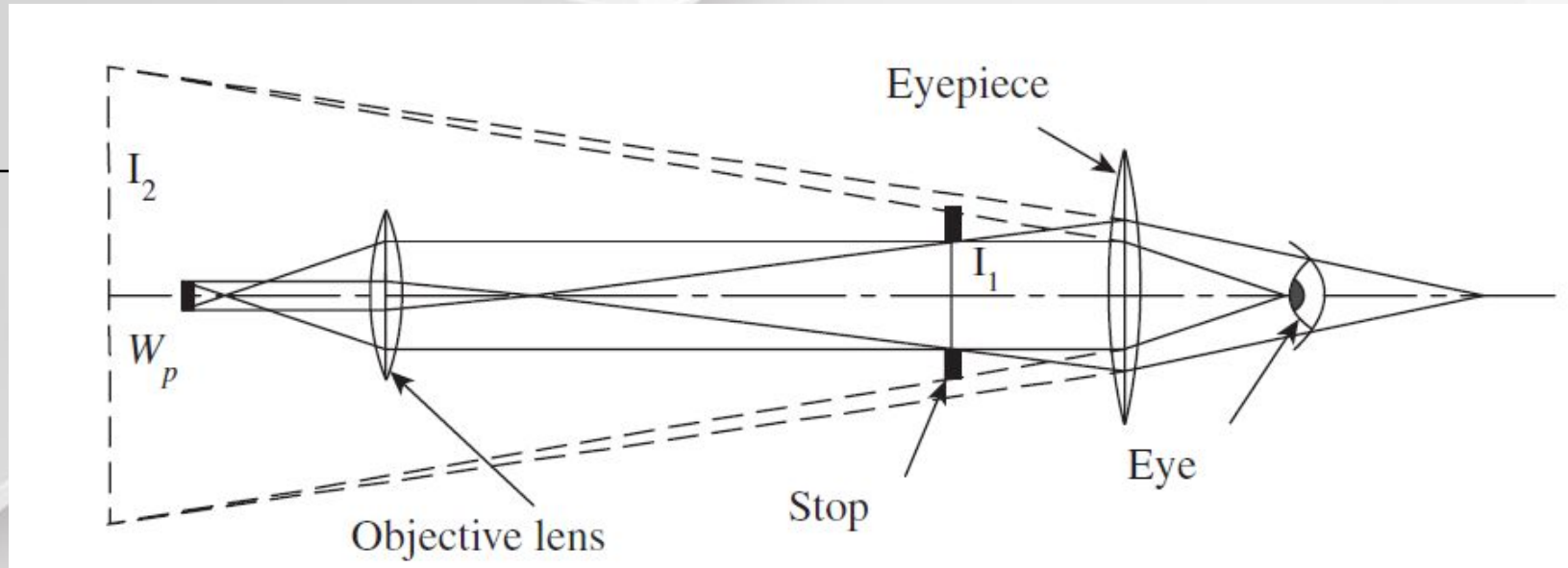
Tool Maker's Microscope

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Principle of microscopy

microscope

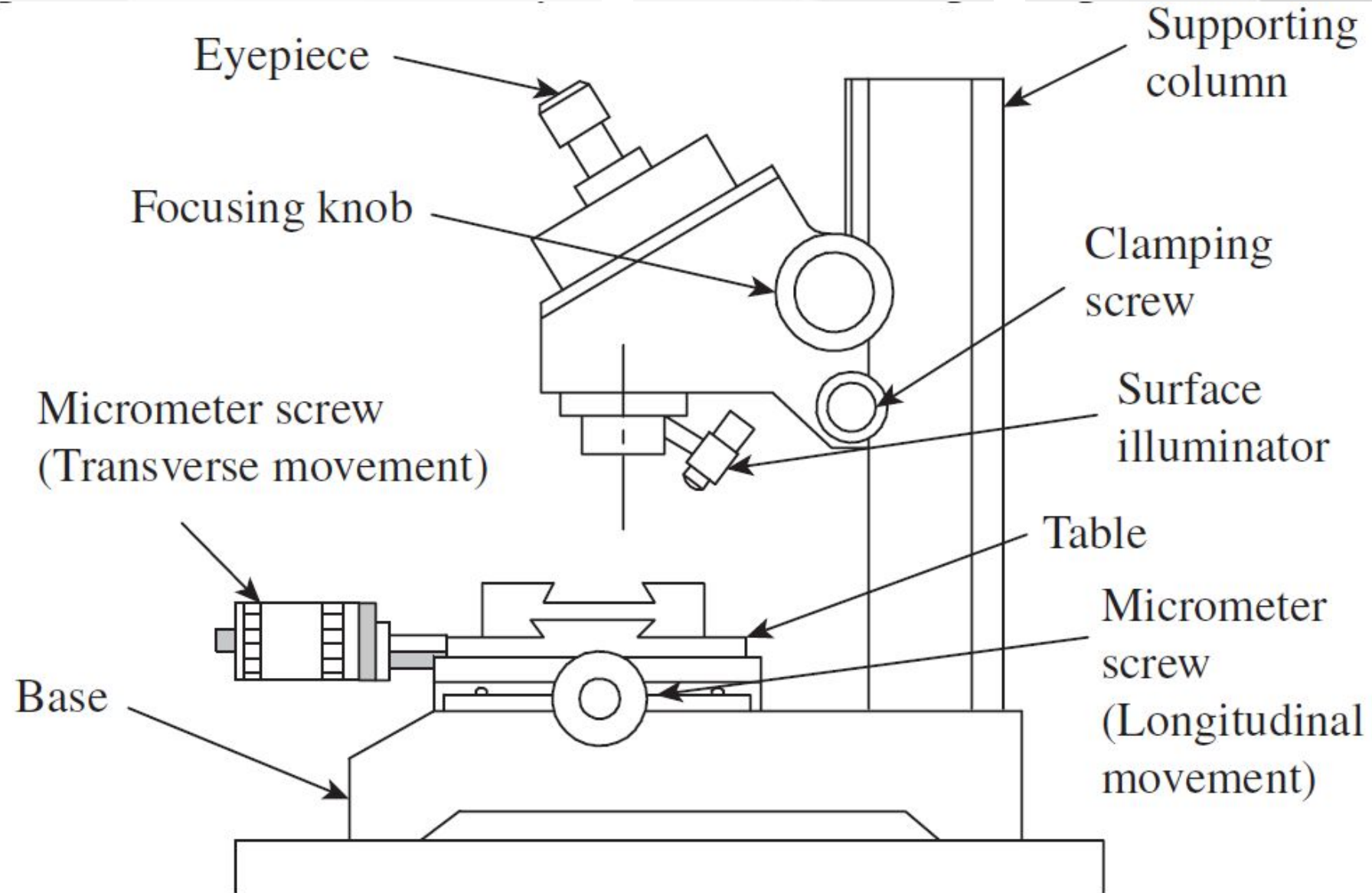
A microscope couples two stages of magnification. The objective lens forms an image of the workpiece at I_1 at the stop. The stop frames the image so that it can be enlarged by the eyepiece. Viewed through the eyepiece, an enlarged virtual image I_2 is obtained. Magnification at each stage multiplies. Thus, a highly effective magnification can be achieved with only moderate magnification at each stage.



Tool Maker's Microscope

It is a multifunctional device that is primarily used for measurement on factory shop floors. Designed with the measurement of workpiece contours and inspection of surface features in mind, a tool maker's microscope supports a wide range of applications from shop floor inspection, and measurement of tools and machined parts to precision measurement of test tools in a measuring room. The main use of a tool maker's microscope is to measure the shape, size, angle, and position of small components that fall under the microscope's measuring range.

Tool maker's microscope

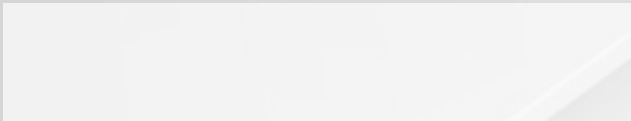


Tool Maker's Microscope

The workpiece is loaded on an XY stage, which has a provision for translatory motion in two principal directions in the horizontal plane. Micrometers are provided for both X and Y axes to facilitate linear measurement to a high degree of accuracy. An angle dial built into the eyepiece portion of the optical tube allows easy angle measurement. A surface illuminator provides the required illumination of the object, so that a sharp and clear image can be obtained.

Tool Maker's Microscope

The element that makes a microscope a measuring instrument is the reticle. When the image is viewed through the eyepiece, the reticle provides a reference or datum to facilitate measurement. Specialized reticles have been developed for precise setting. A typical reticle has two 'crosswires', which can be aligned with a reference line on the image of the workpiece.



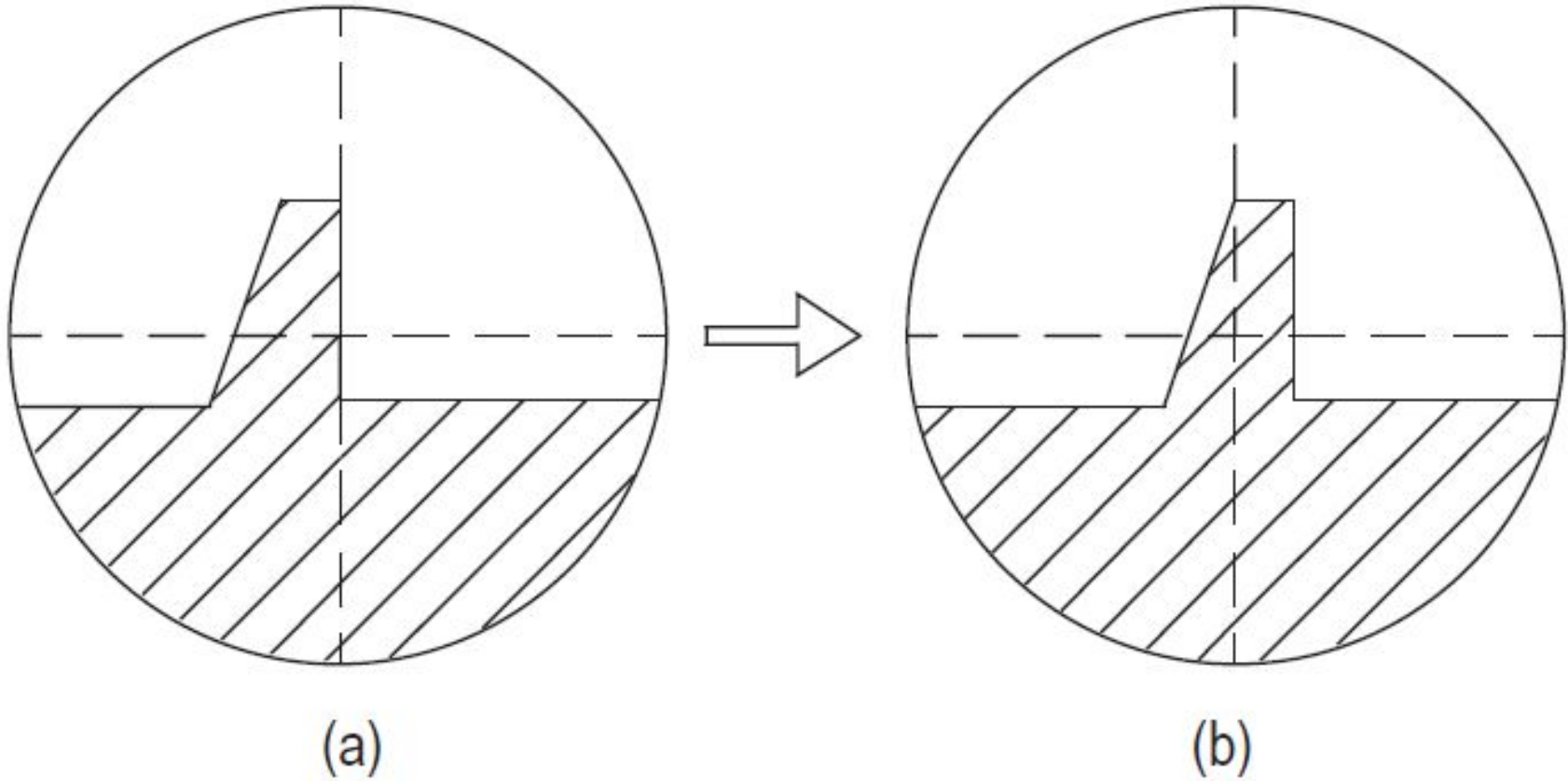


Fig. 7.3 Alignment of cross-wires with the measuring point
(a) Reading R_1 (b) Reading R_2

Tool Maker's Microscope

Applications of Tool Maker's Microscope

1. It is used in shop floor inspection of screw threads, gears, and other small machine parts.
2. Its application includes precision measurement of test tools in tool rooms.
3. It helps determine the dimensions of small holes, which cannot be measured with micrometers and callipers.
4. It facilitates template matching inspection. Small screw threads and involute gear teeth can be inspected using the optional template reticles.
5. It enables inspection of tapers on small components up to an accuracy of 6'.